Node Group and Node Port

What is a Node Group in an EKS Cluster?

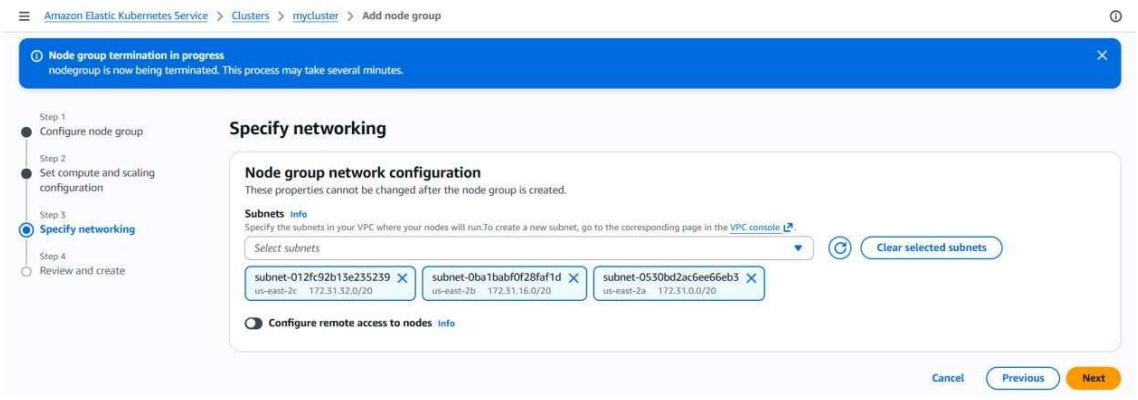
In **Amazon EKS**, a **Node Group** is:

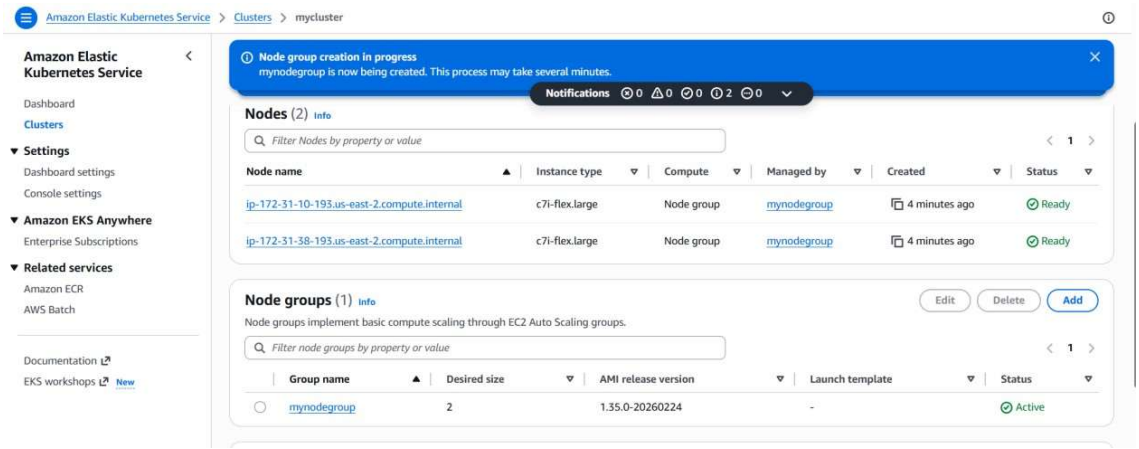
A group of EC2 instances (worker machines) that run your containers (pods).

Simple Understanding

Think like this:

- **Cluster** → Brain (Control Plane)
- **Node Group** → Workers
- **Nodes** → EC2 servers
- **Pods** → Applications running on nodes





- **AWS Account configuration the EC2 Terminal**

```
aws  Search [Alt+S]                                United States (Ohio)  Justin joshva (5117-9408-6556)
                                                                         Justin joshva
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
[ec2-user@ip-172-31-5-128 ~]$ sudo su
[root@ip-172-31-5-128 ec2-user]# aws configure
AWS Access Key ID [None]: AKIAXOKKOAKOCHX36YPS
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-2
Default output format [None]: json
[root@ip-172-31-5-128 ec2-user]# aws s3 ls
[root@ip-172-31-5-128 ec2-user]# aws s3 ls
2026-03-01 06:41:48 justin1234556
[root@ip-172-31-5-128 ec2-user]#
```

What is kubectl?

kubectl is the command-line tool used to interact with a Kubernetes cluster. It allows you to:

- Deploy applications
- Create pods
- Check cluster status • Manage services
- Troubleshoot issues

- **Installation of kubectl**

```
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 57228k 100 57228k   0     0 439.7M     0  --:--:-- --:--:-- --:--:-- 443.5M
[root@ip-172-31-2-11 ec2-user]# kubectl get nodes
NAME                           STATUS   ROLES    AGE   VERSION
ip-172-31-3-190.ec2.internal   Ready    <none>   27m   v1.35.0-eks-efcacff
ip-172-31-87-89.ec2.internal   Ready    <none>   26m   v1.35.0-eks-efcacff
```

Creating the deployement.yaml and service.yaml file

```
[root@ip-172-31-2-11 ec2-user]# kubectl get pod
No resources found in default namespace.
[root@ip-172-31-2-11 ec2-user]# ls
Deployment.yaml  service.yaml
[root@ip-172-31-2-11 ec2-user]# kubectl apply -f Deployment.yaml
deployment.apps/nginx-deployment created
[root@ip-172-31-2-11 ec2-user]# kubectl apply -f service.yaml
service/nginx-nodeport-service created
[root@ip-172-31-2-11 ec2-user]# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-554c4d48d4-2bkb5   1/1     Running   0          30s
nginx-deployment-554c4d48d4-2pwt2   1/1     Running   0          30s
```

Service.yaml (NodePort Example)

If you are using NodePort, use this:

apiVersion: v1 kind: Service metadata:

name: nginx-service spec:

type: NodePort selector: app: nginx ports: - port: 80 targetPort: 80 nodePort: 30007

deployment.yaml

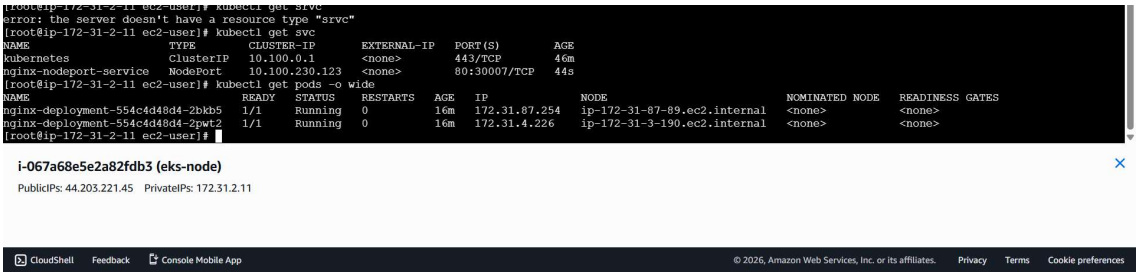
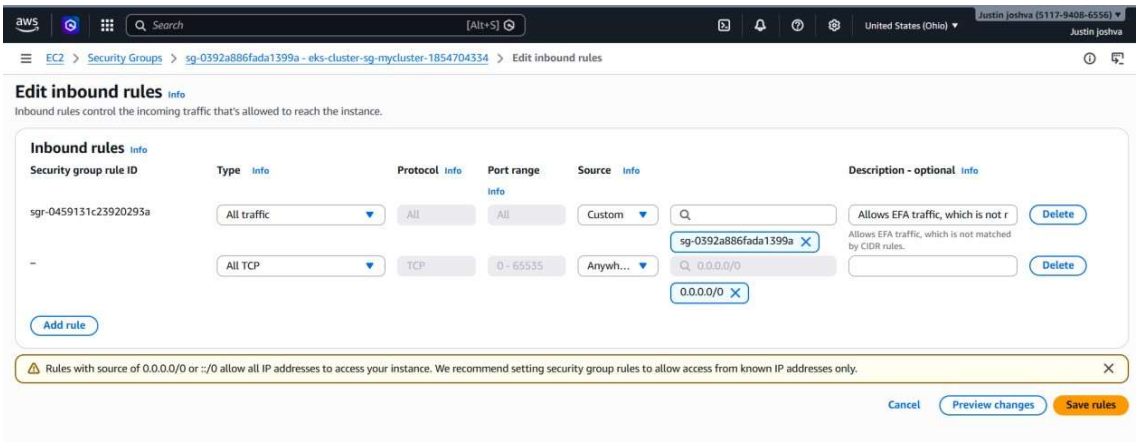
apiVersion: apps/v1 kind: Deployment metadata:

name: nginx-deployment labels: app: nginx spec:

replicas: 2 selector:

matchLabels: app: nginx template: metadata: labels:

app: nginx spec: containers: - name: nginx image: nginx:latest ports:

- containerPort: 80

Created the pods